

and SmartNICs added basic offloading capabilities, the IPU represents a total architectural paradigm shift. It is essentially a specialised computer inside a computer.

A modern IPU typically integrates several key hardware components:

- **High-speed network interfaces:** Capable of handling line-rate speeds of 200Gbps, 400Gbps, or even 800Gbps.
- **Programmable compute cores:** Dedicated processors (often highly efficient ARM-based cores) designed to run the cloud provider's infrastructure software independently of the host CPU.
- **Hardened hardware accelerators:** Custom silicon dedicated to executing highly specific, repetitive tasks at blistering speeds - such as cryptography (IPsec/TLS encryption), data compression, and packet routing.
- **High-bandwidth memory:** Fast, localised memory to prevent bottlenecks during intensive data manipulation.

By placing this powerful, specialised device at the edge of the server motherboard, cloud operators can securely shift the entire burden of a fully virtualised storage and network architecture away from the CPU.

The transformational impact of IPUs in the data centre

The integration of IPUs is not merely an incremental upgrade; it completely redefines how data centres are designed, operated, and monetised. Here are some specific, game-changing advantages that IPUs deliver to modern cloud environments.

Unlocking 100% compute monetisation: The primary financial driver for IPU adoption is CPU reclamation. By offloading the hypervisor, storage management, and networking stack to the IPU, the 'data centre tax' is effectively reduced to zero.

For the cloud service provider, this means that every single core of the expensive, power-hungry server CPU can be dedicated entirely to the tenant's workload. If a user pays for a bare-metal instance or a massive virtual machine, they receive the unhindered, raw performance of the entire CPU. The IPU allows CSPs to achieve maximum density and profitability out of every square foot of their server racks, dramatically lowering the total cost of ownership (TCO).

Bulletproof security and air-tight isolation: In a multi-tenant cloud environment, security is paramount. In a traditional setup, the infrastructure software (the hypervisor) and the tenant's application share the same physical CPU. If a malicious actor successfully compromises the hypervisor or exploits a side-channel vulnerability in the CPU cache, they could potentially access other tenants' data or compromise the entire host system.

The IPU introduces a profound layer of physical and logical separation. Think of a cloud server as a luxury

hotel. You want your high-paying guests (the tenant applications) to have a wonderful experience in their suites. However, you absolutely do not want them having the keys to the boiler room, the security cameras, or the plumbing systems.

With an IPU, the infrastructure control plane is entirely physically separated from the tenant data plane. The cloud provider's security policies, firewalls, and management software run exclusively on the IPU. Even if the tenant completely compromises the host CPU, they cannot breach the underlying network or access the infrastructure management layer, because it is running on a separate processor. This creates an impenetrable root of trust.

Pioneering the diskless data centre: Storage management has traditionally been a rigid and localised problem. Servers have physical hard drives or solid-state drives (SSDs) installed directly within their chassis. This leads to stranded resources: one server might run out of storage while having excess compute, while the server next to it has unused terabytes of storage but a maxed-out processor.

IPUs solve this by enabling the 'diskless data centre'. Through protocols like NVMe-over-Fabrics (NVMe-oF), an IPU can connect a server to a massive, centralised pool of storage located across the network. The IPU manages the complex routing and data compression so efficiently that, to the host CPU, this remote storage behaves as if it is a physical SSD plugged directly into its own motherboard. This disaggregation allows cloud providers to scale compute and storage completely independently, eliminating over-provisioning and drastically reducing hardware waste.

Feeding the AI behemoths: While GPUs and TPUs are the engines of the AI era, they are only as effective as the data pipeline feeding them. AI training requires pulling petabytes of unstructured data from storage and pushing it into the memory of parallel accelerators as fast as possible. If the network or the CPU bottlenecks, the incredibly expensive GPUs end up sitting idle, waiting for data — a phenomenon known as 'data starvation'.

IPUs are the perfect traffic controllers for AI clusters. They handle the East-West network traffic with ultra-low latency, bypass the CPU entirely through Remote Direct Memory Access (RDMA), and ensure that data is encrypted, compressed, and delivered directly to the GPU or TPU memory banks precisely when needed. By keeping the accelerators fed at line rate, IPUs ensure that the multimillion-dollar investments in AI compute are fully utilised.

Comparing architectures: The paradigm shift

To visualise the sheer impact of the IPU, consider how the fundamental architecture of the server changes.